

Avi Garg

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PROFILE

Electronics and Communication Engineering undergraduate with hands-on full-stack development experience across the software development lifecycle – requirements analysis, design, build, and unit testing. Strong analytical skills, attention to detail, and a track record of quickly learning new technologies, tools, and domains.

EDUCATION

Thapar Institute of Engineering and Technology Patiala, Punjab
B.E. in Electronics and Communication Engineering (CGPA: 8.79) Aug 2024 – May 2028

EXPERIENCE

S&T Consultancy Services June 2026 – July 2026
Software Engineering Intern Remote

- **Analysed requirements and performed component-level design** for 8+ RESTful API endpoints using Node.js and Express.js, building and unit testing the code for production applications used by internal and client-facing teams.
- **Debugged and resolved 3+ production issues per sprint**, performing root-cause and impact analysis and implementing fixes supported by unit and integration testing.
- **Reported accurate status on deliverables and flagged risks/issues** to tech leads during Agile ceremonies, contributing to code reviews, CI/CD pipelines, and release activities to deliver features reliably.
- **Shared technical knowledge across the team** through documentation and pair debugging sessions, supporting consistent engineering standards.

PROJECTS

AI Customer Support Platform Personal Project
Next.js, TypeScript, PostgreSQL, ScaleKit, OpenAI API 2026

- **Architected and developed a multi-tenant application** using Next.js and TypeScript, automating customer interactions through the OpenAI API.
- **Designed secure application access controls** with authentication, organization management, and role-based access control, enabling isolated chatbot environments for multiple businesses.
- **Designed and modelled a PostgreSQL database** for business profiles, chatbot configurations, conversation history, and knowledge-base content.

DocForensic Ai Hackathon Project
React, Typescript, Python, FastApi, LangGraph, SQLite 2026

- **Architected a 6-agent LangGraph orchestration pipeline** in Python/FastAPI to analyze invoices, executing 4 specialized agents concurrently to achieve a sub-10 second processing latency per document.
- **Developed a visual forensics engine** with OpenCV and Pillow, leveraging Error Level Analysis and perceptual hashing to detect pixel tampering and cross-document fraud.
- **Integrated a real-time React.js and TypeScript frontend** with the FastAPI backend using Server-Sent Events (SSE), enabling zero-polling live execution traces of the multi-agent DAG.

AI-Powered Delivery Failure Prediction System Hackathon Project
Python, Jupyter Notebook, React, FastAPI, Express, Vercel, Render 2026

- **Performed data analysis and feature engineering** on Amazon delivery data, developing a 61-feature matrix with traffic-pressure, reliability-risk, and distance-cost metrics to identify at-risk orders.
- **Evaluated and tuned an XGBoost classifier** using 5-fold cross-validation, achieving 96.18% validation accuracy, 0.93 F1-score, and 0.99 AUC-ROC.
- **Integrated a React.js frontend with a FastAPI backend** through REST APIs and deployed the application on Vercel and Render for real-time prediction results.

TECHNICAL SKILLS

Languages: C++, JavaScript, TypeScript, SQL, Python
Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, FastAPI, LangChain, Tailwind CSS
Databases & Design: PostgreSQL, MongoDB, Database Design & Modelling
Cloud & DevOps: AWS, Docker, CI/CD, Git/GitHub, Vercel, Render, Postman
Core CS & SDLC: Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Systems, REST API Design, Requirements & Impact Analysis, Unit Testing, Agile/Scrum
Machine Learning: XGBoost, Scikit-learn, Feature Engineering, Model Evaluation

ACHIEVEMENTS

- **Innovation Award NioHack 2026:** Built a finance-focused project as part of Team Nexora, recognized with the Innovation Award at NioHack 2026.
- **Solved 100+ Data Structures Algorithms problems** on LeetCode, strengthening problem-solving and analytical skills.